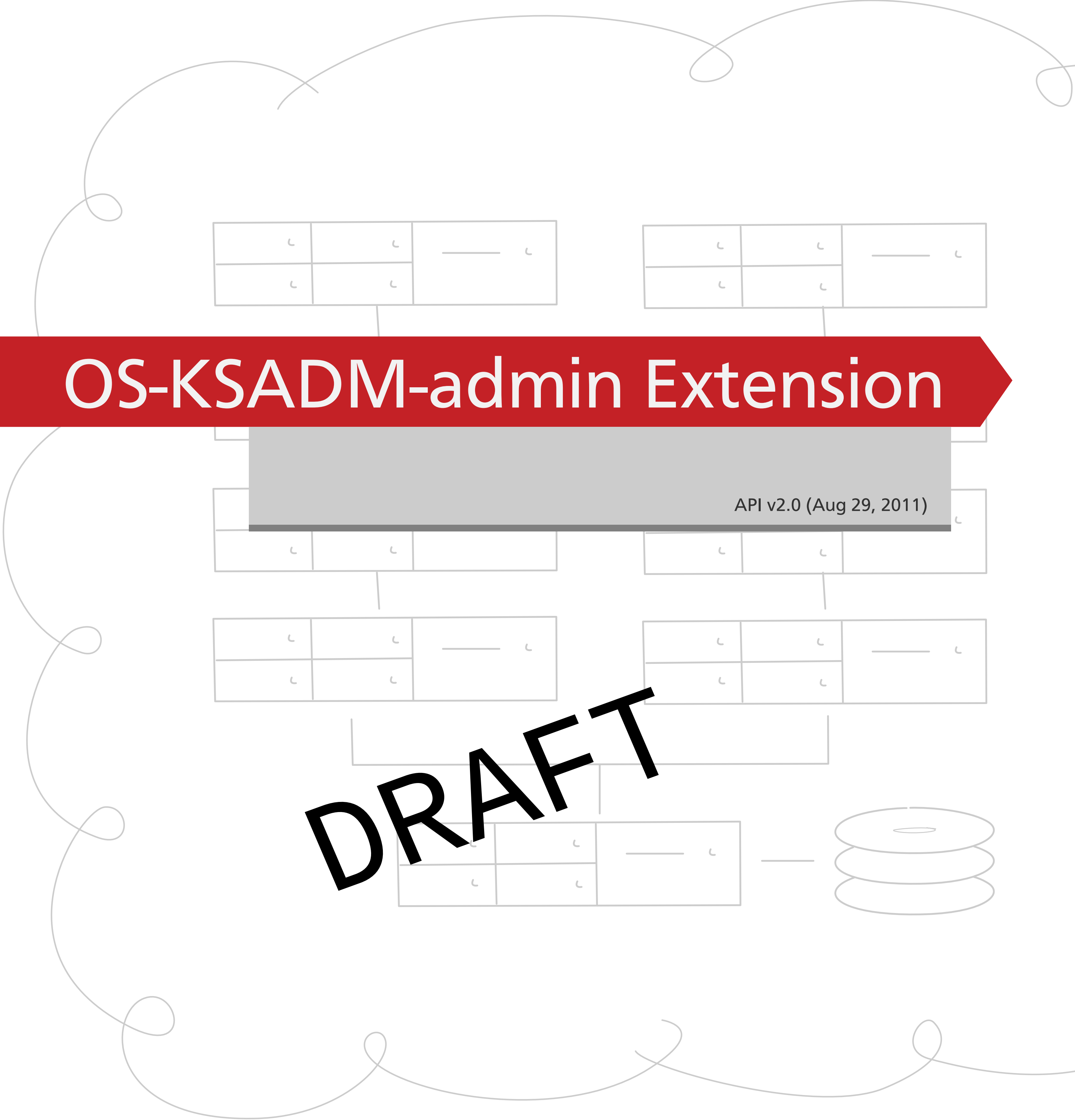

# OS-KSADM-admin Extension

API v2.0 (Aug 29, 2011)

DRAFT

docs.openstack.org

# OS-KSADM-admin Extension

API v2.0 (2011-08-29)
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This document includes details on various operations supported by the OS-KSADM-admin extensions on top of core keystone operations.

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# 1. About This Extension

| | |
|---|---|
| Name | OpenStack KSADM Extension |
| Namespace | http://docs.openstack.org/identity/api/ext/OS-KSADM/v1.0 |
| Alias | OS-KSCATALOG-admin |
| Dependencies | Keystone - OpenStack Identity |
| Doc Link (PDF) | https://github.com/openstack/keystone/raw/master/keystone/content/admin/OS-KSADM-admin-devguide.pdf |
| Doc Link (WADL) | https://raw.github.com/openstack/keystone/master/keystone/content/admin/OS-KSADM-admin.wadl |
| Doc Link (XSD) | https://raw.github.com/openstack/keystone/master/keystone/content/common/xsd/OS-KSADM.xsd |
| Short Description | OpenStack KSADM Admin Extension to Keystone v2.0 API adds the capability to do CRUD on Users, Tenants, Roles and Services. |

## 1.1. Document Change History

| Revision Date | Summary of Changes |
|---|---|
| Sep. 09, 2011 | • Initial release. |

# 2. Extension Operations

| Verb | URI | Description |
|---|---|---|
| User Operations | | |
| **GET** | /users | List users. |
| **POST** | /users | Adds a user. |
| **POST** | /users/{userId} | Update a user. |
| **DELETE** | /users/{userId} | Delete a user. |
| **GET** | /users/{userId}/OS-KSADM/roles?serviceId=*string* | List all the global roles for a user. |
| **PUT** | /users/{userId}/OS-KSADM/roles/{roleId} | Adds a specific global role to a user. |
| **GET** | /users/{userId}/OS-KSADM/roles/{roleId} | Get information about a global role that belongs to a user. |
| **DELETE** | /users/{userId}/OS-KSADM/roles/{roleId} | Deletes a specific global role from a user. |
| **POST** | /users/{userId}/OS-KSADM/credentials | Adds a credential to a user. |
| **GET** | /users/{userId}/OS-KSADM/credentials?marker=*string*&limit=*int* | List credentials. |
| **POST** | /users/{userId}/OS-KSADM/credentials/{credential-type} | Update credentials. |
| **POST** | /users/{userId}/OS-KSADM/credentials/{credential-type} | Delete User credentials. |
| **GET** | /users/{userId}/OS-KSADM/credentials/{credential-type} | Get user credentials. |
| Tenant Operations | | |
| **POST** | /tenants | Creates a tenant. |
| **POST** | /tenants/{tenantId} | Updates a tenant. |
| **DELETE** | /tenants/{tenantId} | Deletes a tenant. |
| **GET** | /tenants/{tenantId}/OS-KSADM/users?marker=*string*&limit=*int* | Lists all the users for a tenant. |
| **GET** | /tenants/{tenantId}/OS-KSADM/users?roleId=*string*&marker=*string*&limit=*int* | Lists all the users with a specific role for a tenant. |
| **GET** | /tenants/{tenantId}/OS-KSADM/roles?marker=*string*&limit=*int* | Lists all the roles for a tenant. |
| **PUT** | /tenants/{tenantId}/OS-KSADM/users/{userId}/roles/{roleId} | Adds a specific role to a user for a tenant. |
| **DELETE** | /tenants/{tenantId}/OS-KSADM/users/{userId}/roles/{roleId} | Deletes a specific role from a user for a tenant. |
| Role Operations | | |
| **GET** | /OS-KSADM/roles?serviceId=*string*&marker=*string*&limit=*int* | List roles. |
| **PUT** | /OS-KSADM/roles | Add a Role. |
| **GET** | /OS-KSADM/roles/{roleId} | Get a role. |
| **DELETE** | /OS-KSADM/roles/{roleId} | Delete a role. |
| Service Operations | | |

| Verb | URI | Description |
|---|---|---|
| **GET** | /OS-KSADM/services?serviceId=*string*&marker=*string*&limit=*int* | List services. |
| **PUT** | /OS-KSADM/services | Add a service. |
| **GET** | /OS-KSADM/services/{serviceId} | Get a service. |
| **DELETE** | /OS-KSADM/services/{serviceId} | Delete a service. |

# 2.1. OS-KSADM Admin Extension API (Service Developer Operations)

Extension operations.

### Table 2.1. Authentication Header

| Header Type | Name | Value |
|---|---|---|
| HTTP/1.1 Request | X-Auth-Token | txfa8426a08eaf |

The following calls are supported by OS-KSADM-admin Extension:

| Verb | URI | Description |
|---|---|---|
| | User Operations | |
| **GET** | /users | List users. |
| **POST** | /users | Adds a user. |
| **POST** | /users/{userId} | Update a user. |
| **DELETE** | /users/{userId} | Delete a user. |
| **GET** | /users/{userId}/OS-KSADM/roles?serviceId=*string* | List all the global roles for a user. |
| **PUT** | /users/{userId}/OS-KSADM/roles/{roleId} | Adds a specific global role to a user. |
| **GET** | /users/{userId}/OS-KSADM/roles/{roleId} | Get information about a global role that belongs to a user. |
| **DELETE** | /users/{userId}/OS-KSADM/roles/{roleId} | Deletes a specific global role from a user. |
| **POST** | /users/{userId}/OS-KSADM/credentials | Adds a credential to a user. |
| **GET** | /users/{userId}/OS-KSADM/credentials?marker=*string*&limit=*int* | List credentials. |
| **POST** | /users/{userId}/OS-KSADM/credentials/{credential-type} | Update credentials. |
| **POST** | /users/{userId}/OS-KSADM/credentials/{credential-type} | Delete User credentials. |
| **GET** | /users/{userId}/OS-KSADM/credentials/{credential-type} | Get user credentials. |
| | Tenant Operations | |
| **POST** | /tenants | Creates a tenant. |
| **POST** | /tenants/{tenantId} | Updates a tenant. |
| **DELETE** | /tenants/{tenantId} | Deletes a tenant. |

| Verb | URI | Description |
|---|---|---|
| **GET** | `/tenants/{tenantId}/OS-KSADM/users?marker=`*string*`&limit=`*int* | Lists all the users for a tenant. |
| **GET** | `/tenants/{tenantId}/OS-KSADM/users?roleId=`*string*`&marker=`*string*`&limit=`*int* | Lists all the users with a specific role for a tenant. |
| **GET** | `/tenants/{tenantId}/OS-KSADM/roles?marker=`*string*`&limit=`*int* | Lists all the roles for a tenant. |
| **PUT** | `/tenants/{tenantId}/OS-KSADM/users/{userId}/roles/{roleId}` | Adds a specific role to a user for a tenant. |
| **DELETE** | `/tenants/{tenantId}/OS-KSADM/users/{userId}/roles/{roleId}` | Deletes a specific role from a user for a tenant. |
| Role Operations | | |
| **GET** | `/OS-KSADM/roles?serviceId=`*string*`&marker=`*string*`&limit=`*int* | List roles. |
| **PUT** | `/OS-KSADM/roles` | Add a Role. |
| **GET** | `/OS-KSADM/roles/{roleId}` | Get a role. |
| **DELETE** | `/OS-KSADM/roles/{roleId}` | Delete a role. |
| Service Operations | | |
| **GET** | `/OS-KSADM/services?serviceId=`*string*`&marker=`*string*`&limit=`*int* | List services. |
| **PUT** | `/OS-KSADM/services` | Add a service. |
| **GET** | `/OS-KSADM/services/{serviceId}` | Get a service. |
| **DELETE** | `/OS-KSADM/services/{serviceId}` | Delete a service. |

### 2.1.1. User Operations

| Verb | URI | Description |
|---|---|---|
| **GET** | `/users` | List users. |
| **POST** | `/users` | Adds a user. |
| **POST** | `/users/{userId}` | Update a user. |
| **DELETE** | `/users/{userId}` | Delete a user. |
| **GET** | `/users/{userId}/OS-KSADM/roles?serviceId=`*string* | List all the global roles for a user. |
| **PUT** | `/users/{userId}/OS-KSADM/roles/{roleId}` | Adds a specific global role to a user. |
| **GET** | `/users/{userId}/OS-KSADM/roles/{roleId}` | Get information about a global role that belongs to a user. |
| **DELETE** | `/users/{userId}/OS-KSADM/roles/{roleId}` | Deletes a specific global role from a user. |
| **POST** | `/users/{userId}/OS-KSADM/credentials` | Adds a credential to a user. |
| **GET** | `/users/{userId}/OS-KSADM/credentials?marker=`*string*`&limit=`*int* | List credentials. |
| **POST** | `/users/{userId}/OS-KSADM/credentials/{credential-type}` | Update credentials. |
| **POST** | `/users/{userId}/OS-KSADM/credentials/{credential-type}` | Delete User credentials. |
| **GET** | `/users/{userId}/OS-KSADM/credentials/{credential-type}` | Get user credentials. |

## 2.1.1.1. List users

| Verb | URI | Description |
|---|---|---|
| **GET** | `/users` | List users. |

List users.

### Table 2.2. List users parameters

| Name | Style | Description |
|---|---|---|
| Request parameters | | |
| `X-Auth-Token` | header | You need a valid admin token for access.<br>String. Required. |

### Normal Response Code(s)

- userss (200 203)

### Error Response Code(s)

- identityFault
- badRequest(400)
- unauthorized(401)
- forbidden(403)
- badMethod(405)
- overLimit(413)
- serviceUnavailable(503)
- itemNotFound(404)

### Example 2.1. List Users Response: XML

```
<?xml version="1.0" encoding="UTF-8"?>
<users xmlns="http://docs.openstack.org/identity/api/v2.0">
    <user xmlns="http://docs.openstack.org/identity/api/v2.0"
          enabled="true" email="john.smith@example.org"
          username="jqsmith" id="u1000"/>
    <user xmlns="http://docs.openstack.org/identity/api/v2.0"
          enabled="true" email="john.smith@example.org"
          username="jqsmith" id="u1001"/>
</users>
```

### Example 2.2. List Users Response: JSON

```
{
    "users":[{
            "id":"u1000",
            "username":"jqsmith",
            "email":"john.smith@example.org",
            "enabled":true
        },
        {
            "id":"u1001",
            "username":"jqsmith",
            "email":"john.smith@example.org",
            "enabled":true
        }
    ],
    "users_links":[]
}
```

### 2.1.1.2. Add user

| Verb | URI | Description |
|---|---|---|
| **POST** | `/users` | Adds a user. |

Adds a user.

#### Table 2.3. Add user parameters

| Name | Style | Description |
|---|---|---|
| Request parameters | | |
| `X-Auth-Token` | header | You need a valid admin token for access.<br>String. Required. |

#### Normal Response Code(s)

- user (201)

#### Error Response Code(s)

- identityFault
- badRequest(400)
- unauthorized(401)
- forbidden(403)
- badMethod(405)
- overLimit(413)
- serviceUnavailable(503)
- itemNotFound(404)
- badMediaType(415)

#### Example 2.3. User Creation Request: XML

```
<?xml version="1.0" encoding="UTF-8"?>
<user xmlns="http://docs.openstack.org/identity/api/v2.0"
      enabled="true" email="john.smith@example.org"
      username="jqsmith"/>
```

#### Example 2.4. User Creation Request: JSON

```
{
  "user": {
    "username": "jqsmith",
    "email": "john.smith@example.org",
    "enabled": true
 }
}
```

#### Example 2.5. User Creation Response: XML

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<user xmlns="http://docs.openstack.org/identity/api/v2.0"
      enabled="true" email="john.smith@example.org"
      username="jqsmith" id="u1000"/>
```

### Example 2.6. User Creation Response: JSON

```
{
  "user": {
    "id": "u1000",
    "username": "jqsmith",
    "email": "john.smith@example.org",
    "enabled": true
 }
}
```

## 2.1.1.3. Update user

| Verb | URI | Description |
|---|---|---|
| **POST** | `/users/{userId}` | Update a user. |

Update a user.

### Table 2.4. Update user parameters

| Name | Style | Description |
|---|---|---|
| Request parameters | | |
| `X-Auth-Token` | header | You need a valid admin token for access.<br>String. Required. |

### Normal Response Code(s)

- user (200)

### Error Response Code(s)

- identityFault
- badRequest(400)
- unauthorized(401)
- forbidden(403)
- badMethod(405)
- overLimit(413)
- serviceUnavailable(503)
- badMediaType(415)
- itemNotFound(404)

### Example 2.7. User Updation Request: XML

```
<?xml version="1.0" encoding="UTF-8"?>
<user xmlns="http://docs.openstack.org/identity/api/v2.0"
      enabled="true" email="john.smith@example.org"
      username="jqsmith" id="u1000"/>
```

### Example 2.8. User Updation Request: JSON

```
{
  "user": {
    "id": "u1000",
    "username": "jqsmith",
    "email": "john.smith@example.org",
    "enabled": true
 }
}
```

### Example 2.9. User Updation Response: XML

```
<?xml version="1.0" encoding="UTF-8"?>
<user xmlns="http://docs.openstack.org/identity/api/v2.0"
      enabled="true" email="john.smith@example.org"
      username="jqsmith" id="u1000"/>
```

### Example 2.10. User Updation Response: JSON

```
{
  "user": {
    "id": "u1000",
    "username": "jqsmith",
    "email": "john.smith@example.org",
    "enabled": true
 }
}
```

## 2.1.1.4. Delete user

| Verb | URI | Description |
|---|---|---|
| **DELETE** | `/users/{userId}` | Delete a user. |

Delete a user.

### Table 2.5. Delete user parameters

| Name | Style | Description |
|---|---|---|
| Request parameters | | |
| `X-Auth-Token` | header | You need a valid admin token for access.<br>String. Required. |

### Normal Response Code(s)

- (204)

### Error Response Code(s)

- identityFault
- badRequest(400)
- unauthorized(401)
- forbidden(403)
- badMethod(405)
- overLimit(413)
- serviceUnavailable(503)
- itemNotFound(404)

### Example 2.11. Delete User Request

```
This operation does not require a request body.
```

## 2.1.1.5. List Global Roles for a User

| Verb | URI | Description |
|---|---|---|
| **GET** | `/users/{userId}/OS-KSADM/roles?serviceId=`*`string`* | List all the global roles for a user. |

List all the global roles for a user.

### Table 2.6. List Global Roles for a User parameters

| Name | Style | Description |
|---|---|---|
| Request parameters | | |
| `X-Auth-Token` | header | You need a valid admin token for access.<br>String. Required. |
| `serviceId` | query | String. Optional. |

### Normal Response Code(s)

- roles (200 203)

### Error Response Code(s)

- identityFault
- badRequest(400)
- unauthorized(401)
- forbidden(403)
- badMethod(405)
- overLimit(413)
- serviceUnavailable(503)
- itemNotFound(404)

### Example 2.12. List Global Roles for User Response: XML

```
<?xml version="1.0" encoding="UTF-8"?>

<roles xmlns="http://docs.openstack.org/identity/api/v2.0">
  <role id="Admin" description="All Access" />
  <role id="Guest" description="Guest Access" />
</roles>
```

### Example 2.13. List Global Roles for User Response: JSON

```
{
    "roles":[{
            "id":"compute:admin"
        }
    ],
    "roles_links":[]
}
```

## 2.1.1.6. Add Global roles to a user.

| Verb | URI | Description |
|---|---|---|
| **PUT** | `/users/{userId}/OS-KSADM/roles/{roleId}` | Adds a specific global role to a user. |

Adds a specific global role to a user.

### Table 2.7. Add Global roles to a user. parameters

| Name | Style | Description |
|---|---|---|
| Request parameters | | |
| `X-Auth-Token` | header | You need a valid admin token for access.<br>String. Required. |
| `roleId` | template | Int. Optional. |

### Normal Response Code(s)

- (200)

### Error Response Code(s)

- identityFault
- badRequest(400)
- unauthorized(401)
- forbidden(403)
- badMethod(405)
- overLimit(413)
- serviceUnavailable(503)
- badMediaType(415)
- itemNotFound(404)

### Example 2.14. Adds global role to a user request

```
This operation does not require a request body.
```

## 2.1.1.7. Get user Role

| Verb | URI | Description |
|---|---|---|
| **GET** | `/users/{userId}/OS-KSADM/roles/{roleId}` | Get information about a global role that belongs to a user. |

Get information about a global role that belongs to a user.

### Table 2.8. Get user Role parameters

| Name | Style | Description |
|---|---|---|
| Request parameters | | |
| `X-Auth-Token` | header | You need a valid admin token for access.<br>String. Required. |
| `roleId` | template | Int. Optional. |

### Normal Response Code(s)

- role (200 203)

### Error Response Code(s)

- identityFault
- badRequest(400)
- unauthorized(401)
- forbidden(403)
- badMethod(405)
- overLimit(413)
- serviceUnavailable(503)
- badMediaType(415)
- itemNotFound(404)

### Example 2.15. List Roles for User Response: XML

```
<?xml version="1.0" encoding="UTF-8"?>

<role xmlns="http://docs.openstack.org/identity/api/v2.0"
  id="Admin"
  description="All Access" />
```

### Example 2.16. List Roles for User Response: JSON

```
{
  "role": {
    "id": "Guest",
    "description": "Guest Access"
  }
}
```

## 2.1.1.8. Delete Global Roles on User.

| Verb | URI | Description |
|---|---|---|
| **DELETE** | `/users/{userId}/OS-KSADM/roles/{roleId}` | Deletes a specific global role from a user. |

Deletes a specific global role from a user.

### Table 2.9. Delete Global Roles on User. parameters

| Name | Style | Description |
|---|---|---|
| Request parameters | | |
| `X-Auth-Token` | header | You need a valid admin token for access.<br>String. Required. |
| `roleId` | template | Int. Optional. |

### Normal Response Code(s)

- (204)

### Error Response Code(s)

- identityFault
- badRequest(400)
- unauthorized(401)
- forbidden(403)
- badMethod(405)
- overLimit(413)
- serviceUnavailable(503)
- itemNotFound(404)

### Example 2.17. Deletes global role from a user

```
This operation does not require a request body.
```

## 2.1.1.9. Add user Credential.

| Verb | URI | Description |
|---|---|---|
| **POST** | `/users/{userId}/OS-KSADM/credentials` | Adds a credential to a user. |

Adds a credential to a user.

### Table 2.10. Add user Credential. parameters

| Name | Style | Description |
|---|---|---|
| Request parameters | | |
| `X-Auth-Token` | header | You need a valid admin token for access.<br>String. Required. |

### Normal Response Code(s)

- credentials (201)

### Error Response Code(s)

- identityFault
- badRequest(400)
- unauthorized(401)
- forbidden(403)
- badMethod(405)
- overLimit(413)
- serviceUnavailable(503)
- badMediaType(415)
- itemNotFound(404)

### Example 2.18. Add Credentials Request: XML

```
<?xml version="1.0" encoding="UTF-8"?>
 <passwordCredentials xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns="http://docs.openstack.org/identity/api/v2.0" username="test_user"
 password="test"/>
```

### Example 2.19. Add Credentials Request: JSON

```
{
    "passwordCredentials": {
        "username": "test_user",
        "password": "mypass"
    }
}
```

### Example 2.20. Add Credential Response: XML

```
<?xml version="1.0" encoding="UTF-8"?>
 <passwordCredentials xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns="http://docs.openstack.org/identity/api/v2.0" username="test_user"
 password="test"/>
```

### Example 2.21. Add Credential Response: JSON

```
{
    "passwordCredentials": {
        "username": "test_user",
        "password": "mypass"
    }
}
```

## 2.1.1.10. List Credentials

| Verb | URI | Description |
|---|---|---|
| **GET** | `/users/{userId}/OS-KSADM/credentials?marker=`*`string`*`&limit=`*`int`* | List credentials. |

List credentials.

### Table 2.11. List Credentials parameters

| Name | Style | Description |
|---|---|---|
| Request parameters | | |
| `X-Auth-Token` | header | You need a valid admin token for access.<br>String. Required. |
| `marker` | query | String. Optional. |
| `limit` | query | Int. Optional. |

### Normal Response Code(s)

- credentials (200 203)

### Error Response Code(s)

- identityFault
- badRequest(400)
- unauthorized(401)
- forbidden(403)
- badMethod(405)

- overLimit(413)
- serviceUnavailable(503)
- itemNotFound(404)

### Example 2.22. List Credentials Response: XML

```
<?xml version="1.0" encoding="UTF-8"?>
<credentials xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xmlns="http://docs.openstack.org/identity/api/v2.0">
    <passwordCredentials username="test_user" password="test"/>
</credentials>
```

### Example 2.23. List Credentials Response: JSON

```
{
    "credentials":[{
            "passwordCredentials":{
                "username":"test_user",
                "password":"mypass"
            }
        }
    ],
    "credentials_links":[]
}
```

## 2.1.1.11. Update user credential

| Verb | URI | Description |
|---|---|---|
| **POST** | `/users/{userId}/OS-KSADM/credentials/{credential-type}` | Update credentials. |

Update credentials.

### Table 2.12. Update user credential parameters

| Name | Style | Description |
|---|---|---|
| | Request parameters | |
| `X-Auth-Token` | header | You need a valid admin token for access.<br>String. Required. |
| `credentialType` | template | ExtensibleCredentialsType. Required. |

### Normal Response Code(s)

- credentials (200)

### Error Response Code(s)

- identityFault
- badRequest(400)
- unauthorized(401)
- forbidden(403)
- badMethod(405)
- overLimit(413)
- serviceUnavailable(503)

- badMediaType(415)
- itemNotFound(404)

### Example 2.24. Update Credential Request: XML

```
<?xml version="1.0" encoding="UTF-8"?>
 <passwordCredentials xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns="http://docs.openstack.org/identity/api/v2.0" username="test_user"
 password="test"/>
```

### Example 2.25. Update Credential Request: JSON

```
{
    "passwordCredentials": {
        "username": "test_user",
        "password": "mypass"
    }
}
```

### Example 2.26. Update Credential Response: XML

```
<?xml version="1.0" encoding="UTF-8"?>
 <passwordCredentials xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns="http://docs.openstack.org/identity/api/v2.0" username="test_user"
 password="test"/>
```

### Example 2.27. Update Credential Response: JSON

```
{
    "passwordCredentials": {
        "username": "test_user",
        "password": "mypass"
    }
}
```

## 2.1.1.12. Delete user credential

| Verb | URI | Description |
|---|---|---|
| **POST** | `/users/{userId}/OS-KSADM/credentials/{credential-type}` | Delete User credentials. |

Delete User credentials.

### Table 2.13. Delete user credential parameters

| Name | Style | Description |
|---|---|---|
| Request parameters | | |
| `X-Auth-Token` | header | You need a valid admin token for access.<br>String. Required. |
| `credentialType` | template | ExtensibleCredentialsType. Required. |

### Normal Response Code(s)

- (204)

### Error Response Code(s)

- identityFault
- badRequest(400)
- unauthorized(401)
- forbidden(403)
- badMethod(405)
- overLimit(413)
- serviceUnavailable(503)
- badMediaType(415)
- itemNotFound(404)

### Example 2.28. Delete User Credentials

```
This operation does not require a request body.
```

## 2.1.1.13. Get user Credentials

| Verb | URI | Description |
|---|---|---|
| **GET** | `/users/{userId}/OS-KSADM/credentials/{credential-type}` | Get user credentials. |

Get user credentials.

### Table 2.14. Get user Credentials parameters

| Name | Style | Description |
|---|---|---|
| Request parameters | | |
| `X-Auth-Token` | header | You need a valid admin token for access.<br>String. Required. |
| `credentialType` | template | ExtensibleCredentialsType. Required. |

### Normal Response Code(s)

- credentials (200 203)

### Error Response Code(s)

- identityFault
- badRequest(400)
- unauthorized(401)
- forbidden(403)
- badMethod(405)
- overLimit(413)
- serviceUnavailable(503)
- itemNotFound(404)

### Example 2.29. Get User Credential Response: XML

```
<?xml version="1.0" encoding="UTF-8"?>
 <passwordCredentials xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
 xmlns="http://docs.openstack.org/identity/api/v2.0" username="test_user"
password="test"/>
```

### Example 2.30. Get User Credential Response: JSON

```
{
    "passwordCredentials": {
        "username": "test_user",
        "password": "mypass"
    }
}
```

## 2.1.2. Tenant Operations

| Verb | URI | Description |
|---|---|---|
| **POST** | `/tenants` | Creates a tenant. |
| **POST** | `/tenants/{tenantId}` | Updates a tenant. |
| **DELETE** | `/tenants/{tenantId}` | Deletes a tenant. |
| **GET** | /tenants/{tenantId}/OS-KSADM/users?marker=*string*&limit=*int* | Lists all the users for a tenant. |
| **GET** | /tenants/{tenantId}/OS-KSADM/users?roleId=*string*&marker=*string*&limit=*int* | Lists all the users with a specific role for a tenant. |
| **GET** | /tenants/{tenantId}/OS-KSADM/roles?marker=*string*&limit=*int* | Lists all the roles for a tenant. |
| **PUT** | `/tenants/{tenantId}/OS-KSADM/users/{userId}/roles/{roleId}` | Adds a specific role to a user for a tenant. |
| **DELETE** | `/tenants/{tenantId}/OS-KSADM/users/{userId}/roles/{roleId}` | Deletes a specific role from a user for a tenant. |

### 2.1.2.1. Add Tenant

| Verb | URI | Description |
|---|---|---|
| **POST** | `/tenants` | Creates a tenant. |

This call creates a tenant.

### Table 2.15. Add Tenant parameters

| Name | Style | Description |
|---|---|---|
| | Request parameters | |
| `X-Auth-Token` | header | You need a valid admin token for access.<br>String. Required. |

### Normal Response Code(s)

- tenant (201)

### Error Response Code(s)

- identityFault
- badRequest(400)

- unauthorized(401)
- forbidden(403)
- badMethod(405)
- overLimit(413)
- serviceUnavailable(503)
- badMediaType(415)

### Example 2.31. Tenant Creation Request: XML

```
<?xml version="1.0" encoding="UTF-8"?>
<tenant xmlns="http://docs.openstack.org/identity/api/v2.0"
        enabled="true" name="ACME Corp">
    <description>A description...</description>
</tenant>
```

### Example 2.32. Tenant Creation Request: JSON

```
{
  "tenant": {
    "name": "ACME corp",
    "description": "A description ...",
    "enabled": true
  }
}
```

### Example 2.33. Tenant Creation Response: XML

```
<?xml version="1.0" encoding="UTF-8"?>
<tenant xmlns="http://docs.openstack.org/identity/api/v2.0"
        enabled="true" id="1234" name="ACME Corp">
    <description>A description...</description>
</tenant>
```

### Example 2.34. Tenant Creation Response: JSON

```
{
  "tenant": {
    "id": "1234",
    "name": "ACME corp",
    "description": "A description ...",
    "enabled": true
  }
}
```

## 2.1.2.2. Update Tenant

| Verb | URI | Description |
|---|---|---|
| **POST** | `/tenants/{tenantId}` | Updates a tenant. |

This call updates a tenant.

### Table 2.16. Update Tenant parameters

| Name | Style | Description |
|---|---|---|
| Request parameters | | |
| `X-Auth-Token` | header | You need a valid admin token for access. |

| Name | Style | Description |
|---|---|---|
| | | String. Required. |
| `tenantId` | template | String. Optional. |

## Normal Response Code(s)

- tenant (200)

## Error Response Code(s)

- identityFault
- badRequest(400)
- unauthorized(401)
- forbidden(403)
- badMethod(405)
- overLimit(413)
- serviceUnavailable(503)
- itemNotFound(404)
- badMediaType(415)

## Example 2.35. Tenant Update Request: XML

```
<?xml version="1.0" encoding="UTF-8"?>
<tenant xmlns="http://docs.openstack.org/identity/api/v2.0"
        enabled="true" id="1234" name="ACME Corp">
    <description>A description...</description>
</tenant>
```

## Example 2.36. Tenant Update Request: JSON

```
{
  "tenant": {
    "id": "1234",
    "name": "ACME corp",
    "description": "A description ...",
    "enabled": true
  }
}
```

## Example 2.37. Tenant Update Response: XML

```
<?xml version="1.0" encoding="UTF-8"?>
<tenant xmlns="http://docs.openstack.org/identity/api/v2.0"
        enabled="true" id="1234" name="ACME Corp">
    <description>A description...</description>
</tenant>
```

## Example 2.38. Tenant Update Response: JSON

```
{
  "tenant": {
    "id": "1234",
    "name": "ACME corp",
    "description": "A description ...",
    "enabled": true
  }
}
```

## 2.1.2.3. Delete a Tenant

| Verb | URI | Description |
|---|---|---|
| **DELETE** | `/tenants/{tenantId}` | Deletes a tenant. |

This call deletes a tenant.

### Table 2.17. Delete a Tenant parameters

| Name | Style | Description |
|---|---|---|
| | Request parameters | |
| `X-Auth-Token` | header | You need a valid admin token for access.<br>String. Required. |
| `tenantId` | template | String. Optional. |

### Normal Response Code(s)

- (204)

### Error Response Code(s)

- identityFault
- badRequest(400)
- unauthorized(401)
- forbidden(403)
- badMethod(405)
- overLimit(413)
- serviceUnavailable(503)
- itemNotFound(404)

### Example 2.39. Delete Tenant Request

```
This operation does not require a request body.
```

## 2.1.2.4. List users for a Tenant.

| Verb | URI | Description |
|---|---|---|
| **GET** | `/tenants/{tenantId}/OS-KSADM/users ?marker=`*`string`*`&limit=`*`int`* | Lists all the users for a tenant. |

Lists all the users for a tenant.

### Table 2.18. List users for a Tenant. parameters

| Name | Style | Description |
|---|---|---|
| | Request parameters | |
| `X-Auth-Token` | header | You need a valid admin token for access.<br>String. Required. |
| `marker` | query | String. Optional. |

| Name | Style | Description |
|---|---|---|
| `limit` | query | Int. Optional. |
| `tenantId` | template | String. Optional. |

### Normal Response Code(s)

- users (200 203)

### Error Response Code(s)

- identityFault
- badRequest(400)
- unauthorized(401)
- forbidden(403)
- badMethod(405)
- overLimit(413)
- serviceUnavailable(503)
- itemNotFound(404)

### Example 2.40. List Users for Tenant Response: XML

```
<?xml version="1.0" encoding="UTF-8"?>
<users xmlns="http://docs.openstack.org/identity/api/v2.0">
    <user xmlns="http://docs.openstack.org/identity/api/v2.0"
          enabled="true" email="john.smith@example.org"
          username="jqsmith" id="u1000"/>
    <user xmlns="http://docs.openstack.org/identity/api/v2.0"
          enabled="true" email="john.smith@example.org"
          username="jqsmith" id="u1001"/>
</users>
```

### Example 2.41. List Users for Tenant Response: JSON

```
{
    "users":[{
            "id":"u1000",
            "username":"jqsmith",
            "email":"john.smith@example.org",
            "enabled":true
        },
        {
            "id":"u1001",
            "username":"jqsmith",
            "email":"john.smith@example.org",
            "enabled":true
        }
    ],
    "users_links":[]
}
```

## 2.1.2.5. List user with role for tenant.

| Verb | URI | Description |
|---|---|---|
| **GET** | `/tenants/{tenantId}/OS-KSADM/users?roleId=`*string*`&marker=`*string*`&limit=`*int* | Lists all the users with a specific role for a tenant. |

Lists all the users with a specific role for a tenant.

### Table 2.19. List user with role for tenant. parameters

| Name | Style | Description |
|---|---|---|
| Request parameters | | |
| `X-Auth-Token` | header | You need a valid admin token for access.<br>String. Required. |
| `roleId` | query | String. Required. |
| `marker` | query | String. Optional. |
| `limit` | query | Int. Optional. |
| `tenantId` | template | String. Optional. |

## Normal Response Code(s)

- users (200 203)

## Error Response Code(s)

- identityFault
- badRequest(400)
- unauthorized(401)
- forbidden(403)
- badMethod(405)
- overLimit(413)
- serviceUnavailable(503)
- itemNotFound(404)

### Example 2.42. List Users with a specific role for a Tenant Response: XML

```
<?xml version="1.0" encoding="UTF-8"?>
<users xmlns="http://docs.openstack.org/identity/api/v2.0">
    <user xmlns="http://docs.openstack.org/identity/api/v2.0"
          enabled="true" email="john.smith@example.org"
          username="jqsmith" id="u1000"/>
    <user xmlns="http://docs.openstack.org/identity/api/v2.0"
          enabled="true" email="john.smith@example.org"
          username="jqsmith" id="u1001"/>
</users>
```

### Example 2.43. List Users with a specific role for a Tenant Response: JSON

```
{
    "users":[{
            "id":"u1000",
            "username":"jqsmith",
            "email":"john.smith@example.org",
            "enabled":true
        },
        {
            "id":"u1001",
            "username":"jqsmith",
            "email":"john.smith@example.org",
            "enabled":true
        }
    ],
    "users_links":[]
}
```

## 2.1.2.6. List roles for a Tenant.

| Verb | URI | Description |
|---|---|---|
| **GET** | `/tenants/{tenantId}/OS-KSADM/roles ?marker=`*`string`*`&limit=`*`int`* | Lists all the roles for a tenant. |

Lists all the roles for a tenant.

### Table 2.20. List roles for a Tenant. parameters

| Name | Style | Description |
|---|---|---|
| | | Request parameters |
| `X-Auth-Token` | header | You need a valid admin token for access.<br>String. Required. |
| `marker` | query | String. Optional. |
| `limit` | query | Int. Optional. |
| `tenantId` | template | String. Optional. |

### Normal Response Code(s)

- roles (200 203)

### Error Response Code(s)

- identityFault
- badRequest(400)
- unauthorized(401)
- forbidden(403)
- badMethod(405)
- overLimit(413)
- serviceUnavailable(503)
- itemNotFound(404)

### Example 2.44. List Roles for Tenant Response: XML

```
<?xml version="1.0" encoding="UTF-8"?>

<roles xmlns="http://docs.openstack.org/identity/api/v2.0">
  <role id="Admin" description="All Access" />
  <role id="Guest" description="Guest Access" />
</roles>
```

### Example 2.45. List Roles for Tenant Response: JSON

```
{
    "roles":[{
            "id":"compute:admin"
        }
    ],
    "roles_links":[]
}
```

## 2.1.2.7. Add roles on a tenant.

| Verb | URI | Description |
|---|---|---|
| **PUT** | `/tenants/{tenantId}/OS-KSADM/users/{userId}/roles/{roleId}` | Adds a specific role to a user for a tenant. |

Adds a specific role to a user for a tenant.

### Table 2.21. Add roles on a tenant. parameters

| Name | Style | Description |
|---|---|---|
| Request parameters | | |
| `X-Auth-Token` | header | You need a valid admin token for access.<br>String. Required. |
| `tenantId` | template | String. Optional. |
| `userId` | template | String. Optional. |

### Normal Response Code(s)

- (200)

### Error Response Code(s)

- identityFault
- badRequest(400)
- unauthorized(401)
- forbidden(403)
- badMethod(405)
- overLimit(413)
- serviceUnavailable(503)
- badMediaType(415)
- itemNotFound(404)

### Example 2.46. Add role to a user for a Tenant Request

```
This operation does not require a request body.
```

## 2.1.2.8. Delete roles on tenant.

| Verb | URI | Description |
|---|---|---|
| **DELETE** | `/tenants/{tenantId}/OS-KSADM/users/{userId}/roles/{roleId}` | Deletes a specific role from a user for a tenant. |

Deletes a specific role from a user for a tenant.

### Table 2.22. Delete roles on tenant. parameters

| Name | Style | Description |
|---|---|---|
| Request parameters | | |

| Name | Style | Description |
|---|---|---|
| `X-Auth-Token` | header | You need a valid admin token for access.<br>String. Required. |
| `tenantId` | template | String. Optional. |
| `userId` | template | String. Optional. |

### Normal Response Code(s)

- (204)

### Error Response Code(s)

- identityFault
- badRequest(400)
- unauthorized(401)
- forbidden(403)
- badMethod(405)
- overLimit(413)
- serviceUnavailable(503)
- itemNotFound(404)

### Example 2.47. Delete Role From User For a Tenant Request

```
This operation does not require a request body.
```

## 2.1.3. Role Operations

| Verb | URI | Description |
|---|---|---|
| **GET** | `/OS-KSADM/roles?serviceId=`*`string`*`&marker=`*`string`*`&limit=`*`int`* | List roles. |
| **PUT** | `/OS-KSADM/roles` | Add a Role. |
| **GET** | `/OS-KSADM/roles/{roleId}` | Get a role. |
| **DELETE** | `/OS-KSADM/roles/{roleId}` | Delete a role. |

### 2.1.3.1. List Roles

| Verb | URI | Description |
|---|---|---|
| **GET** | `/OS-KSADM/roles?serviceId=`*`string`*`&marker=`*`string`*`&limit=`*`int`* | List roles. |

List roles.

### Table 2.23. List Roles parameters

| Name | Style | Description |
|---|---|---|
| Request parameters | | |
| `X-Auth-Token` | header | You need a valid admin token for access.<br>String. Required. |

| Name | Style | Description |
|---|---|---|
| `serviceId` | query | String. Optional. |
| `marker` | query | String. Optional. |
| `limit` | query | Int. Optional. |

### Normal Response Code(s)

- roles (200 203)

### Error Response Code(s)

- identityFault
- badRequest(400)
- unauthorized(401)
- forbidden(403)
- badMethod(405)
- overLimit(413)
- serviceUnavailable(503)
- itemNotFound(404)

### Example 2.48. List Roles Response: XML

```
<?xml version="1.0" encoding="UTF-8"?>

<roles xmlns="http://docs.openstack.org/identity/api/v2.0">
  <role id="Admin" description="All Access" />
  <role id="Guest" description="Guest Access" />
</roles>
```

### Example 2.49. List Roles Response: JSON

```
{
    "roles":[{
            "id":"compute:admin"
        }
    ],
    "roles_links":[]
}
```

## 2.1.3.2. Add Role

| Verb | URI | Description |
|---|---|---|
| **PUT** | `/OS-KSADM/roles` | Add a Role. |

Add a Role.

### Table 2.24. Add Role parameters

| Name | Style | Description |
|---|---|---|
| | Request parameters | |
| `X-Auth-Token` | header | You need a valid admin token for access.<br>String. Required. |

### Normal Response Code(s)

- role (201)

### Error Response Code(s)

- identityFault
- badRequest(400)
- unauthorized(401)
- forbidden(403)
- badMethod(405)
- overLimit(413)
- serviceUnavailable(503)
- badMediaType(415)
- itemNotFound(404)

### Example 2.50. Add a Role Request: XML

```
<?xml version="1.0" encoding="UTF-8"?>

<role xmlns="http://docs.openstack.org/identity/api/v2.0"
  id="Admin"
  description="All Access" />
```

### Example 2.51. Add a Role Request: JSON

```
{
  "role": {
    "id": "Guest",
    "description": "Guest Access"
  }
}
```

### Example 2.52. Add a Role Response: XML

```
<?xml version="1.0" encoding="UTF-8"?>

<role xmlns="http://docs.openstack.org/identity/api/v2.0"
  id="Admin"
  description="All Access" />
```

### Example 2.53. Add a Role Response: JSON

```
{
  "role": {
    "id": "Guest",
    "description": "Guest Access"
  }
}
```

## 2.1.3.3. Get Role

| Verb | URI | Description |
|---|---|---|
| **GET** | `/OS-KSADM/roles/{roleId}` | Get a role. |

Get a role.

### Table 2.25. Get Role parameters

| Name | Style | Description |
|---|---|---|
| Request parameters | | |
| `X-Auth-Token` | header | You need a valid admin token for access.<br>String. Required. |
| `roleId` | template | String. Optional. |

### Normal Response Code(s)

- role (200 203)

### Error Response Code(s)

- identityFault
- badRequest(400)
- unauthorized(401)
- forbidden(403)
- badMethod(405)
- overLimit(413)
- serviceUnavailable(503)
- badMediaType(415)
- itemNotFound(404)

### Example 2.54. Get a role: XML

```
<?xml version="1.0" encoding="UTF-8"?>

<role xmlns="http://docs.openstack.org/identity/api/v2.0"
  id="Admin"
  description="All Access" />
```

### Example 2.55. Get a role: JSON

```
{
  "role": {
    "id": "Guest",
    "description": "Guest Access"
  }
}
```

## 2.1.3.4. Delete Role

| Verb | URI | Description |
|---|---|---|
| **DELETE** | `/OS-KSADM/roles/{roleId}` | Delete a role. |

Delete a role.

### Table 2.26. Delete Role parameters

| Name | Style | Description |
|---|---|---|
| Request parameters | | |
| `X-Auth-Token` | header | You need a valid admin token for access. |

| Name | Style | Description |
|---|---|---|
| | | String. Required. |
| `roleId` | template | String. Optional. |

### Normal Response Code(s)

- (204)

### Error Response Code(s)

- identityFault
- badRequest(400)
- unauthorized(401)
- forbidden(403)
- badMethod(405)
- overLimit(413)
- serviceUnavailable(503)
- itemNotFound(404)

### Example 2.56. Delete a Role

```
This operation does not require a request body.
```

## 2.1.4. Service Operations

| Verb | URI | Description |
|---|---|---|
| **GET** | `/OS-KSADM/services?serviceId=`*`string`*`&marker=`*`string`*`&limit=`*`int`* | List services. |
| **PUT** | `/OS-KSADM/services` | Add a service. |
| **GET** | `/OS-KSADM/services/{serviceId}` | Get a service. |
| **DELETE** | `/OS-KSADM/services/{serviceId}` | Delete a service. |

### 2.1.4.1. List Services

| Verb | URI | Description |
|---|---|---|
| **GET** | `/OS-KSADM/services?serviceId=`*`string`*`&marker=`*`string`*`&limit=`*`int`* | List services. |

List services.

### Table 2.27. List Services parameters

| Name | Style | Description |
|---|---|---|
| Request parameters | | |
| `X-Auth-Token` | header | You need a valid admin token for access.<br>String. Required. |
| `serviceId` | query | String. Optional. |
| `marker` | query | String. Optional. |

| Name | Style | Description |
|---|---|---|
| `limit` | query | Int. Optional. |

### Normal Response Code(s)

- services (200 203)

### Error Response Code(s)

- identityFault
- badRequest(400)
- unauthorized(401)
- forbidden(403)
- badMethod(405)
- overLimit(413)
- serviceUnavailable(503)
- itemNotFound(404)

### Example 2.57. List services Response: XML

```
<?xml version="1.0" encoding="UTF-8"?>

<services xmlns="http://docs.openstack.org/identity/api/ext/OS-KSADM/v1.0">
  <service id="Nova" type="compute" description="Openstack Service" />
  <service id="Glance" type="image-service" description="Openstack Service" />
</services>
```

### Example 2.58. List services Response: JSON

```
{
    "OS-KSADM:services":[{
            "id":"nova",
            "type":"compute",
            "description":"Computers in the Cloud"
        },
        {
            "id":"glance",
            "type":"image-service",
            "description":"Server Images"
        }
    ],
    "OS-KSADM:services_links":[]
}
```

## 2.1.4.2. Add Service

| Verb | URI | Description |
|---|---|---|
| **PUT** | `/OS-KSADM/services` | Add a service. |

Add a service.

### Table 2.28. Add Service parameters

| Name | Style | Description |
|---|---|---|
| Request parameters | | |

| Name | Style | Description |
|---|---|---|
| `X-Auth-Token` | header | You need a valid admin token for access.<br>String. Required. |

### Normal Response Code(s)

- service (201)

### Error Response Code(s)

- identityFault
- badRequest(400)
- unauthorized(401)
- forbidden(403)
- badMethod(405)
- overLimit(413)
- serviceUnavailable(503)
- badMediaType(415)
- itemNotFound(404)

### Example 2.59. Add a service Request: XML

```
<?xml version="1.0" encoding="UTF-8"?>

<service xmlns="http://docs.openstack.org/identity/api/ext/OS-KSADM/v1.0" id=
"nova"
 type="compute" description="OpenStack Compute Service" />
```

### Example 2.60. Add a service Request: JSON

```
{
    "OS-KSADM:service":{
        "id":"nova",
        "type":"compute",
        "description":"A description ..."
    }
}
```

### Example 2.61. Add a service Response: XML

```
<?xml version="1.0" encoding="UTF-8"?>

<service xmlns="http://docs.openstack.org/identity/api/ext/OS-KSADM/v1.0" id=
"nova"
 type="compute" description="OpenStack Compute Service" />
```

### Example 2.62. Add a service Response: JSON

```
{
    "OS-KSADM:service":{
        "id":"nova",
        "type":"compute",
        "description":"A description ..."
    }
}
```

## 2.1.4.3. Get Service

| Verb | URI | Description |
|---|---|---|
| **GET** | `/OS-KSADM/services/{serviceId}` | Get a service. |

Get a service.

### Table 2.29. Get Service parameters

| Name | Style | Description |
|---|---|---|
| Request parameters | | |
| `X-Auth-Token` | header | You need a valid admin token for access.<br>String. Required. |
| `serviceId` | template | Int. Optional. |

### Normal Response Code(s)

- service (200 203)

### Error Response Code(s)

- identityFault
- badRequest(400)
- unauthorized(401)
- forbidden(403)
- badMethod(405)
- overLimit(413)
- serviceUnavailable(503)
- itemNotFound(404)

### Example 2.63. Get a service Response: XML

```
<?xml version="1.0" encoding="UTF-8"?>

<service xmlns="http://docs.openstack.org/identity/api/ext/OS-KSADM/v1.0" id=
"nova"
 type="compute" description="OpenStack Compute Service" />
```

### Example 2.64. Get a service Response: JSON

```
{
    "OS-KSADM:service":{
        "id":"nova",
        "type":"compute",
        "description":"A description ..."
    }
}
```

## 2.1.4.4. Delete Service

| Verb | URI | Description |
|---|---|---|
| **DELETE** | `/OS-KSADM/services/{serviceId}` | Delete a service. |

Delete a service.

### Table 2.30. Delete Service parameters

| Name | Style | Description |
|---|---|---|
| Request parameters | | |
| `X-Auth-Token` | header | You need a valid admin token for access.<br>String. Required. |
| `serviceId` | template | Int. Optional. |

### Normal Response Code(s)

- (204)

### Error Response Code(s)

- identityFault
- badRequest(400)
- unauthorized(401)
- forbidden(403)
- badMethod(405)
- overLimit(413)
- serviceUnavailable(503)
- itemNotFound(404)

### Example 2.65. Delete a Service

```
This operation does not require a request body.
```